

From Static Embeddings to Dynamic World Models: Navigating the Baseline Paradox in Single-Cell Foundation Models

Single-cell genomics is undergoing a paradigm shift from task-specific architectures to large-scale foundation models capable of learning universal cellular representations. However, this transition faces a critical "baseline paradox," where parameter-free linear representations often outperform complex deep learning models on static downstream benchmarks, challenging the necessity of massive scale for descriptive tasks [103]. This review synthesizes recent advancements addressing this dichotomy by pivoting from static embedding to dynamic, generative "world models." We examine how frameworks like Lingshu-Cell and scDiVa utilize masked discrete diffusion to simulate counterfactual cellular states and perturbation responses, capabilities inaccessible to static baselines [32]. Furthermore, we explore the integration of multimodal data and biological priors, such as gene regulatory networks and natural language, to enhance interpretability and causal reasoning beyond statistical co-expression [4]. By contrasting the efficiency of simplified architectures with the predictive power of semi-mechanistic simulators, this document highlights the emerging consensus that future progress relies on hybrid systems. These systems must balance computational scalability with mechanistic fidelity to achieve robust, interpretable, and universally applicable tools for virtual cell development and drug discovery.

Introduction: The Paradigm Shift to Single-Cell Foundation Models

The field of single-cell genomics is currently undergoing a fundamental transformation, characterized by a decisive move away from task-specific deep learning architectures and static clustering methods toward large-scale foundation models capable of learning universal, transferable representations of cellular states. Historically, the analysis of single-cell RNA sequencing (scRNA-seq) data relied heavily on conventional machine learning approaches that struggled with the high dimensionality, sparsity, and heterogeneous dependency structures inherent in these datasets [26]. While early deep learning solutions offered superior performance over traditional methods for tasks such as imputation, clustering, and cell-type annotation, they often lacked the generalizability required to synthesize knowledge across diverse biological contexts [66]. The emergence of foundation models, inspired by breakthroughs in natural language processing (NLP), addresses these limitations by treating genes as tokens and cells as sentences, thereby capturing complex, high-dimensional dependencies through self-supervised learning on massive, unlabeled corpora [38]. This paradigm shift leverages the transformer architecture to model the "language of life," enabling the learning of context-dependent gene interactions that mirror syntactic and semantic structures in human language [73].

However, this transition is not merely a scaling exercise; it is accompanied by a critical re-evaluation of the necessity and interpretability of complex models compared to simpler baselines. A growing body of evidence suggests that much of the biologically relevant structure in current benchmarks may already be accessible through low-complexity, parameter-free linear representations, challenging the assumption that massive deep learning models are strictly required for state-of-the-art performance [103]. This "baseline paradox" has spurred a dual track of research: one pushing the boundaries of scalable architectures like state space models (SSMs) to handle long genomic sequences efficiently, and another focusing on rigorous benchmarking to distinguish genuine biological insight from artifacts of data manifold structure [16; 178]. Consequently, the field is moving toward models that explicitly integrate external biological knowledge, such as gene regulatory networks and cell ontologies, to enhance expressiveness and reduce computational overhead while ensuring biological fidelity [21; 4].

The evolution of these models also reflects a growing recognition of the need to incorporate multimodal data and natural language to overcome the limitations of purely transcriptomic views. Purely transcriptomic models often neglect the rich semantic information available in scientific literature and cell ontologies, leading to innovations like LangCell and scMMGPT, which unify single-cell data with natural language descriptions to enable zero-shot cell identity understanding and improved generalization in out-of-distribution scenarios [57; 24]. Similarly, the integration of spatial context and proteomic data expands the scope of these models, as seen in frameworks that utilize hierarchical graph transformers to capture microenvironmental influences and panel-agnostic protein expressions [81; 141]. Furthermore, the "black box" nature of deep learning has driven the development of interpretability techniques, such as sparse autoencoders and transcoders, which extract biologically meaningful concepts and decision circuits from latent spaces, bridging the gap between correlation and causality [121; 41].

Moreover, the field is increasingly moving toward generative "world models" capable of simulating cellular responses to perturbations, a capability essential for virtual cell development and drug discovery. Models like Lingshu-Cell and SAVE employ masked discrete diffusion and conditional transformers to generate realistic transcriptomic profiles and predict responses to unseen perturbations, moving beyond static representation learning to dynamic simulation [32; 34]. However, the efficacy of large language models (LLMs) in directly reasoning about cell biology remains a subject of rigorous evaluation, with benchmarks revealing that while generalist LLMs show preliminary understanding, they often fall short of specialist models in complex mechanistic tasks [30; 172]. Consequently, hybrid approaches that combine the reasoning capabilities of LLMs with the specialized representation power of single-cell foundation models are emerging as promising directions to harness the synergies between textual knowledge and omics data [49; 171]. This convergence of scalable architecture, multimodal integration, and generative capability defines the current frontier of single-cell foundation models, setting the stage for more robust, interpretable, and universally applicable tools in computational biology.

The Evolution from Static Embeddings to Dynamic World Models

Early computational pipelines in single-cell biology predominantly treated cells as independent samples, relying on static embeddings to characterize cellular identity. While effective for clustering and annotation, these approaches often failed to capture the contextual nuances, distributional relationships, and stochastic nature inherent in gene expression data. Traditional methods reduced high-dimensional transcriptomic profiles to fixed vector representations, assuming a cell's state could be fully described by its position in a latent space at a single time point [103]. However, this static paradigm overlooks the dynamic reality of biological systems, where cells exist in continuous flux driven by internal regulatory programs and external perturbations. The limitation of static models is particularly evident when attempting to predict cellular responses to interventions, as they cannot inherently simulate the trajectory of state transitions or the variability of outcomes across a population [73]. Furthermore, recent benchmarks have revealed a "baseline paradox," where simple, parameter-free linear representations often outperform complex foundation models on static downstream tasks, suggesting that current evaluation metrics may not adequately capture the biological complexity that deep learning aims to model [103]. This paradox underscores the need for a paradigm shift from descriptive cataloging to predictive simulation, where the true value of foundation models lies in their ability to generate counterfactual states rather than merely embedding observed ones.

Recent advancements have marked a transition toward generative "world models" and dynamic representations that explicitly simulate cellular state transitions. This evolution is exemplified by frameworks like Lingshu-Cell, which operates as a generative cellular world model using a masked discrete diffusion architecture. Unlike previous foundation models optimized for static representation learning, Lingshu-Cell learns the distribution of transcriptomic states, enabling the conditional simulation of cells under various perturbations without relying on arbitrary gene selection or continuous noise assumptions that misalign with sparse count data [32]. By modeling the intrinsic state space directly, such systems move beyond descriptive cataloging to create *in silico* environments capable of high-fidelity simulation, analogous to world models in artificial intelligence that support conditional reasoning and planning. Similarly, scDiVa employs a masked discrete diffusion foundation model that aligns generation with the dropout-like corruption process, utilizing a bidirectional denoiser to jointly model discrete gene identities and continuous values, thereby preserving global cell identity while addressing the sparsity of single-cell data [43]. These approaches recognize that gene expression is fundamentally discrete and unordered, requiring generative processes that respect the exchangeability of genes rather than imposing artificial sequential dependencies.

The transition from static to dynamic modeling also involves a fundamental rethinking of how gene regulatory networks (GRNs) and temporal dynamics are inferred. Early dynamical models often relied on ordinary differential equations (ODEs) that assumed deterministic behavior or added *ad hoc* noise, failing to capture the "bursty" nature of transcription revealed by single-cell data. In contrast, newer mechanistic approaches like Harissa utilize stochastic dynamical models driven by transcriptional bursting to infer network structures that naturally account for cell-to-cell variability without external noise terms [154]. Furthermore, methods such as scPN have emerged to simultaneously infer pseudotime, velocity fields, and gene interaction networks using piecewise ODEs, addressing the inconsistencies found in traditional pipelines that compute velocity and pseudotime sequentially [277]. These developments highlight a growing consensus that accurate modeling requires integrating mechanistic insights with data-driven learning to resolve the complex, non-linear dynamics of cell fate decisions. Complementary approaches like FLeCS incorporate gene network structure into coupled differential equations to model transcriptional dynamics at scale, avoiding low-dimensional projections to provide functional insights into perturbed mechanisms [59].

A critical aspect of this evolution is the move from predicting mean expression changes to modeling full distributional shifts. Traditional perturbation models often assumed a single fixed response distribution conditioned on cell type and perturbation, ignoring the manifold of possible outcomes driven by unobserved latent factors. PerturbDiff addresses this by shifting the modeling focus from individual cells to entire distributions, employing a diffusion-based process over probability distributions in a Hilbert space to capture population-level response shifts [45]. Similarly, scDFM utilizes conditional flow matching to model the full distribution of perturbed cells, incorporating a Perturbation-Aware Differential Transformer to capture context-specific expression changes and improve robustness against sparsity [19]. These distributional approaches are essential for capturing the stochasticity of biological responses, allowing for the simulation of heterogeneous outcomes that static mean-prediction models cannot reproduce [149]. Other frameworks,

such as Departures, approximate Schrödinger Bridges to directly align control and perturbed distributions, capturing single-cell heterogeneity through joint training of discrete and continuous models [Departures: Distributional Transport for Single-Cell Perturbation Prediction with Neural Schrodinger Bridges].

The concept of the "virtual cell" has further expanded to include white-box simulators and causal frameworks that offer interpretability alongside predictive power. VCWorld integrates structured biological knowledge with large language models to create a cell-level white-box simulator that generates interpretable, stepwise predictions of signaling cascades, addressing the "black box" limitations of purely data-driven models [237]. Concurrently, latent causal diffusion models like LCD frame gene expression as a stationary diffusion process to disentangle measurement noise from biological signal, enabling the recovery of causal structures through causal linearization [174]. This integration of causal inference with generative modeling represents a significant leap forward, allowing researchers to not only predict unseen perturbation effects but also map the underlying regulatory mechanisms driving them. Moreover, the geometry of cellular state space is increasingly understood through the lens of dynamic transitions rather than static clusters. Studies on cell phenotypic transitions have revealed that cells often traverse concerted mechanisms involving increased frustration in gene regulatory networks before settling into new stable states, a phenomenon that static embeddings fail to capture [219]. To model these transitions, frameworks like Chreode employ structured residual transition operators to predict action-conditioned state transitions in a single step, effectively shifting distributional evolution from inference time to training time [137].

Despite these advances, challenges remain in scaling these dynamic models and ensuring their biological fidelity. While some studies suggest that simple linear representations can outperform complex foundation models on certain static benchmarks, the capacity to simulate dynamic, counterfactual scenarios remains the exclusive domain of sophisticated generative world models [103]. The field is thus converging on a hybrid approach that combines the scalability of foundation models with the mechanistic rigor of dynamical systems, aiming to create universal models capable of generalizing across diverse tissues, species, and perturbation contexts [76]. This synthesis of static representation learning and dynamic simulation constitutes the core of the emerging paradigm in single-cell foundation models, paving the way for truly predictive and interpretable virtual biology.

Challenges in Data Heterogeneity, Sparsity, and Standardization

The transition toward single-cell foundation models represents a paradigm shift in computational biology, yet this progress is fundamentally constrained by the intrinsic statistical properties of single-cell RNA sequencing (scRNA-seq) data. The most pervasive obstacle is extreme data sparsity, characterized by dropout rates often exceeding 90%, where true biological expression is recorded as zero due to technical limitations [276]. This sparsity complicates the distinction between biological silence and technical artifact, leading many reconstruction-based models to inadvertently encode measurement noise rather than stable cellular programs [1]. While traditional imputation methods attempt to recover latent signals through smoothing or matrix factorization, comprehensive benchmarks suggest that deep learning-based imputation does not consistently outperform simpler statistical approaches and may even distort biological variability essential for downstream clustering and trajectory analysis [106]. Consequently, the field is increasingly moving toward architectures that are robust to sparsity by design, such as masked discrete diffusion models that treat dropouts as part of the corruption process rather than errors to be corrected [43], or latent predictive architectures that shift learning from reconstructing sparse counts to predicting in latent space [1].

Beyond sparsity, batch effects and technical heterogeneity pose significant barriers to creating universal models. Single-cell datasets are frequently generated across diverse laboratories, protocols, and conditions, introducing non-biological variations that can obscure true cellular states [280]. Although foundation models aim to learn batch-invariant representations, their performance often degrades when faced with covariate shifts or unseen donors, particularly in organ-specific contexts where cellular architecture is highly dynamic [46]. The challenge is compounded by the lack of standardized schemas for representing data, especially in emerging modalities like scATAC-seq, where the high dimensionality and binary nature of chromatin accessibility data require specialized tokenization strategies distinct from transcriptomics [119]. Without rigorous standardization, models trained on aggregated atlases risk learning dataset-specific artifacts rather than generalizable biological principles. This concern is starkly highlighted by studies showing that simple, parameter-free linear representations can sometimes outperform complex foundation models on out-of-distribution tasks, suggesting that much of the biologically relevant structure in current benchmarks is accessible through low-complexity linear representations [103]. This "baseline paradox" implies that model capacity alone is insufficient without architectural innovations tailored to the unique statistical structure of single-cell omics, particularly when data quality and quantity fall short of textual standards [67]. Indeed, efficient alternatives grounded in optimal transport on Gaussian mixtures with fewer than 500k parameters have demonstrated comparable results to massive foundation models, reinforcing the need for streamlined approaches in data-limited regimes [199].

A critical and often under-addressed dimension of heterogeneity is the long-tailed distribution of cell types, where rare but biologically crucial populations are severely underrepresented. This imbalance leads to models that excel at annotating common cell types while failing to capture the nuances of rare subtypes, which are frequently drivers of disease progression or therapeutic resistance [56]. Standard training objectives, which minimize global reconstruction error, tend to prioritize the majority class, effectively suppressing the signal from rare populations. To mitigate this, recent innovations include specialized loss functions like Gaussian Inflation Loss and hard data mining strategies that dynamically re-weight samples to force the model to attend to minority classes [56]. Furthermore, the integration of prior biological knowledge, such as cell ontology graphs, has been proposed to guide representation learning, ensuring that the structural relationships between rare and common cell types are preserved even when data is scarce [60].

The complexity of these data challenges is further exacerbated in perturbation modeling, where the goal is to predict cellular responses to genetic or chemical interventions. Here, the "perturbation suppression hypothesis" suggests that dominant, perturbation-invariant background signals often overwhelm the sparse, specific signals induced by interventions, causing foundation models to fail in generalizing to unseen perturbations [185]. Additionally, the destructive nature of sequencing means that control and perturbed states are unpaired, requiring models to learn distributional mappings rather than direct cell-to-cell transitions, a task complicated by the inherent heterogeneity of cellular responses within the same condition [45]. Addressing these multifaceted issues requires a departure from one-size-fits-all architectures toward specialized frameworks that explicitly model data sparsity, correct for batch effects without erasing biological variation, and account for the heavy-tailed nature of cellular diversity. While scaling laws indicate that performance improves with data volume, the data-limited regimes typical of rare cell types or specific

perturbation conditions suggest that model capacity alone is insufficient without architectural innovations tailored to the unique statistical structure of single-cell omics [67].

Architectural Innovations in Single-Cell Foundation Models

The architectural landscape of single-cell foundation models has undergone a rapid transformation, driven by the need to balance computational scalability with the biological fidelity required to model complex transcriptomic data. Initially, the field was dominated by transformer-based architectures that adapted natural language processing paradigms to genomics, treating genes as tokens and cells as sentences. While models such as those surveyed in [38] demonstrated that attention mechanisms could effectively learn contextual dependencies for tasks like cell type annotation, the inherent quadratic complexity of standard self-attention imposed a severe scalability ceiling. This limitation renders pure transformer architectures computationally prohibitive for modeling long sequences characteristic of whole transcriptomes involving tens of thousands of genes, a constraint explicitly noted in [16]. Consequently, the field has pivoted toward State Space Models (SSMs), which offer linear-time processing capabilities while maintaining the ability to model long-range dependencies essential for gene regulatory interactions. Architectures like GeneMamba incorporate BiMamba modules to efficiently capture gene context information, enabling the scalable processing of over 50 million cells while significantly reducing computational costs compared to traditional transformers. Similarly, [178] introduces a patch-based cell tokenization strategy built on state space duality, allowing for the integration of multi-omics data without the aggressive feature selection that often discards crucial biological signals.

To balance the trade-off between local expressive power and global efficiency, hybrid architectures have emerged as a promising middle ground. [48] proposes a modular design combining a transformer-based local-context encoder with an SSM-based global sequential decoder. This architecture preserves the expressive power of transformers for capturing complex, order-independent gene interactions while leveraging the linear efficiency of Mamba for global integration, effectively addressing the computational overhead that limits pure attention mechanisms. Parallel to these efficiency-driven innovations, there has been a fundamental rethinking of generative objectives to better align with the discrete, sparse, and non-sequential nature of scRNA-seq data. Early generative attempts often relied on continuous diffusion models or autoregressive frameworks that imposed artificial ordering biases or misaligned noise corruption strategies. In response, masked discrete diffusion models have emerged as a biologically coherent alternative. [32] introduces a framework that operates directly in discrete token space, utilizing a masking-and-prediction objective to learn transcriptomic state distributions without relying on prior gene selection. This approach allows for non-autoregressive, bidirectional refinement of whole-transcriptome profiles, accurately reproducing cellular heterogeneity across diverse tissues. Complementing this, [43] aligns the generation process with the dropout-like corruption inherent in single-cell data, employing a bidirectional denoiser that jointly models discrete gene identities and continuous values. These models effectively simulate cellular states and predict perturbation responses by respecting the exchangeability of gene expression data, a principle further reinforced by [Scalable Single-Cell Gene Expression Generation with Latent Diffusion Models], which utilizes unified Multi-head Cross-Attention Blocks to respect the permutation-invariant nature of gene profiles.

Despite these architectural advancements, a growing body of evidence suggests that standard reconstruction objectives in deep learning models often fail to capture latent regulatory signals, leading to the "Baseline Paradox" where parameter-free linear representations outperform complex foundation models on downstream benchmarks [103]. Systematic evaluations reveal that attention patterns in models like scGPT and Geneformer often encode structured biological information such as protein-protein interactions but provide no incremental value for perturbation prediction compared to trivial gene-level baselines [8]. Furthermore, sparse autoencoder analyses indicate that while these models internalize organized biological knowledge like pathway membership, they capture minimal causal regulatory logic [6]. To address this disconnect between statistical co-expression and mechanistic causality, recent architectural innovations increasingly integrate explicit biological priors and structural constraints. Frameworks like [4] systematically integrate multi-scale Gene Regulatory Networks (GRNs) into the training process, employing graph topological adapters and edge perturbation strategies to address information asymmetry and dynamically weight regulatory relationships. Similarly, [21] proposes a constraint-based approach to connect foundation models with gene network graphs, expanding receptive fields without causing parameter explosion by focusing on learning predefined graph structures. These efforts reflect a broader consensus that future architectural innovations must move beyond statistical co-expression to incorporate the mechanistic rigor of biological network dynamics, thereby bridging the gap between deep learning representational power and causal regulatory logic.

Transformer Adaptations and Efficiency Strategies

The application of standard transformer architectures to single-cell and genomic data is frequently hindered by computational bottlenecks arising from the quadratic complexity of softmax attention and the inherent instability of training on high-dimensional, sparse biological sequences. To overcome these limitations, recent research has pivoted toward architectural innovations that fundamentally rethink the attention mechanism and encoder-decoder structures. A primary avenue of improvement involves replacing the standard softmax function with sigmoid attention, which addresses the gradient explosion issues often encountered in long biological sequences. [159] demonstrates that sigmoid attention offers globally bounded derivatives and a diagonal Jacobian structure, eliminating the need for gradient clipping and preventing catastrophic divergence during training. Empirically, this modification yields a 25% improvement in cell-type separation and accelerates training speeds by up to 10% compared to softmax counterparts, supported by optimized GPU kernels like TritonSigmoid that achieve high throughput on modern hardware. This shift suggests that stability and efficiency in biological foundation models may require moving away from the dense coupling inherent in softmax attention.

Parallel to modifications in attention functions, structural asymmetries in encoder-decoder designs have emerged as a critical strategy for managing the sparsity and dimensionality of single-cell RNA sequencing (scRNA-seq) data. Classical symmetric transformers often struggle with the memory and computation requirements of datasets containing millions of cells and tens of thousands of genes. [99] introduces an asymmetric encoder-decoder architecture specifically tailored to leverage the sparse characteristics of scRNA-seq data. By decoupling the encoding and decoding processes, this approach reduces floating-point operations (FLOPs) by one to two orders of magnitude compared to classical transformers, enabling the training of massive models on the largest available datasets without sacrificing accuracy in downstream tasks like cell type annotation and drug combination prediction. This design philosophy aligns with broader efforts to scale foundation models efficiently, as evidenced by [67], which establishes that clear power-law scaling behaviors emerge in single-cell genomics only when sufficient data is available to match model capacity, highlighting the necessity of efficient architectures to reach these regimes.

Beyond attention mechanism substitution and asymmetric designs, the integration of State Space Models (SSMs) represents a paradigm shift toward linear-time complexity for long-range dependency modeling. While transformers excel at short-context tasks, their inability to extrapolate to sequences longer than those seen during training limits their utility in genomics, where regulatory elements can be separated by vast genomic distances. [223] demonstrates that SSM-inspired architectures, such as Caduceus and Hawk, can match transformer performance while exhibiting impressive zero-shot extrapolation capabilities across contexts 10 to 100 times longer than training sequences. Similarly, [48] adopts a hybrid approach, combining transformer-based local context encoders with SSM-based global decoders to capture both intricate gene interactions and long-range regulatory signals with linear efficiency. This hybridization addresses the limitations of pure transformer models, which incur $\mathcal{O}(n^2)$ complexity, and pure SSM models, which may require fixed input ordering, offering a balanced solution for transcriptome modeling. Furthermore, [178] extends this logic to multi-omics, utilizing patch-based tokenization and state space duality to integrate data without discarding crucial biological information through aggressive feature selection.

The drive for efficiency also extends to the handling of full-length data and the mitigation of information loss. Traditional preprocessing often relies on selecting highly variable genes, potentially discarding crucial biological signals. [58] counters this by employing a bidirectional Hyena operator within a transformer framework, allowing for the processing of full-length scRNA-seq data without truncation or aggressive feature selection. This approach preserves the complete information content of raw data, enhancing accuracy in tasks such as cell type classification and imputation. Additionally, the challenge of batch size limitations in contrastive learning, often exacerbated by high-dimensional data, is addressed in [200]. This work introduces a divide-and-conquer strategy that decouples batch size from GPU memory constraints, enabling the training of large-scale models like CellLM on millions of cells and achieving state-of-the-art performance in drug sensitivity prediction.

Despite these architectural advances, the field remains cognizant of the trade-offs between complexity and necessity. [103] presents a compelling counterpoint, suggesting that for many current benchmarks, simple linear representations derived from careful normalization can match or exceed the performance of computationally intensive foundation models. This finding implies that while architectural innovations like sigmoid attention and asymmetric encoders are

vital for scaling and stability, the biological signal in many datasets may be accessible through simpler means, urging the community to rigorously evaluate whether increased model complexity translates to genuine biological insight or merely overfitting to dataset artifacts. Nevertheless, for tasks requiring the modeling of long-range dependencies, massive scale, or generative capabilities, the adapted transformer architectures and hybrid SSM approaches described above provide the necessary computational foundation to push the boundaries of single-cell and genomic analysis.

Generative Frameworks: Diffusion, Flow Matching, and Optimal Transport

The transition from static representation learning to dynamic generative modeling represents a fundamental paradigm shift in single-cell foundation models, driven primarily by the destructive nature of sequencing technologies which precludes longitudinal tracking of individual cells. Because the same cell cannot be observed before and after a perturbation, modern architectures must abandon the assumption of one-to-one cell correspondences and instead model the transport of entire probability distributions. This necessitates frameworks capable of mapping unpaired control and perturbed populations while accounting for the inherent heterogeneity of cellular responses, which often form a manifold of possible distributions rather than a single fixed outcome. To address this, recent advancements have coalesced around three principled mathematical foundations: conditional flow matching, diffusion processes, and optimal transport, each offering distinct advantages in capturing population-level shifts and latent biological variability.

Flow matching has emerged as a robust strategy for defining continuous, time-dependent transformations between unperturbed and perturbed states, offering scalability and stability in high-dimensional gene expression spaces. SAVE introduces a unified generative framework that leverages conditional Transformers with a coarse-grained representation, grouping semantically related genes into blocks to capture higher-order dependencies often missed by token-independent models [34]. By integrating a Flow Matching mechanism with a condition-masking strategy, SAVE enables flexible simulation and generalization to unseen condition combinations, particularly excelling in low-resource settings where data scarcity is a bottleneck. Similarly, scDFM utilizes conditional flow matching to model the full distribution of perturbed cells, incorporating a Maximum Mean Discrepancy (MMD) objective to align populations beyond individual cell correspondences [19]. This approach is enhanced by a Perturbation-Aware Differential Transformer architecture that leverages gene interaction graphs, demonstrating significant improvements in combinatorial perturbation settings where traditional methods often fail due to the complexity of interacting effects. Extending this logic across diverse cellular contexts, CFM-GP employs a continuous, time-dependent transformation conditioned on cell type, allowing a single model to predict transcriptional effects across multiple cell types without requiring cell type-specific training [52]. These flow-based methods collectively underscore the advantage of modeling distributional shifts directly, offering superior robustness to noise and sparsity compared to approaches assuming fixed response distributions. Furthermore, scBIG explicitly models coordinated gene programs through module-inductive representations, inducing coherent gene programs from data via Gene-Relation Clustering before applying conditional flow matching to enable flexible and generalizable perturbation prediction [17].

Parallel to flow matching, diffusion models have been adapted to handle the unique statistical properties of transcriptomic data, with a notable divergence between continuous and discrete formulations. PerturbDiff addresses the variability in cellular responses by embedding distributions as points in a Hilbert space, defining a diffusion-based generative process that operates directly over probability distributions rather than individual cells [45]. This allows the model to capture population-level response shifts across hidden latent factors, generalizing substantially better to unseen perturbations. In contrast to continuous assumptions, Lingshu-Cell introduces a masked discrete diffusion model that operates directly in a discrete token space, aligning with the sparse and non-sequential nature of single-cell data [32]. By avoiding the artificial ordering bias of autoregressive models and the global continuous-noise corruption of standard denoising diffusion probabilistic models, Lingshu-Cell accurately simulates transcriptomic distributions across diverse species and tissues. Similarly, scDiVa employs a masked discrete diffusion foundation model with a continuous-time forward masking mechanism in token space, jointly modeling discrete gene identities and continuous values to preserve global cell identity [43]. These discrete approaches highlight a growing consensus that generative paradigms must be tailored to the count-based, dropout-prone nature of genomic data to avoid artifacts introduced by continuous relaxations. Additionally, StateXDiff introduces a cell State-contextualized multimodal diffusion framework that integrates transcriptomic profiles with inferred protein features to predict drug-induced cellular state changes, addressing out-of-distribution conditions through a Virtual Multimodal Cell State [74].

Optimal Transport (OT) provides another principled mathematical foundation for handling unpaired data, offering rigorous guarantees for mapping between distributions while preserving geometric structure. The Conditional Monge Gap learns OT maps conditionally on arbitrary covariates, such as drug dosage or cell type, enabling cross-task learning that unlocks remarkable generalization abilities to unseen treatments [120]. By aggregating data across conditions, this

method outperforms condition-specific models in capturing higher moments of the perturbed population distribution, effectively modeling the heterogeneity of cellular responses. Furthermore, Departures approximates Schrödinger Bridges to define stochastic dynamic mappings that recover entropy-regularized optimal transport, directly aligning control and perturbed distributions across different conditions without relying on bidirectional modeling [71]. This approach leverages MinibatchOT based pairing to guide bridge learning, effectively capturing single-cell heterogeneity in both discrete gene activation states and continuous expression distributions. Unlasting also utilizes Dual Diffusion Implicit Bridges (DDIB) to learn mappings between unpaired data distributions, interpreting this process as a form of data augmentation that incorporates gene regulatory network information to propagate perturbation signals biologically [35].

The integration of biological priors further distinguishes these modern generative frameworks, moving them beyond purely data-driven black boxes toward mechanistic interpretability. STRAND conditions perturbation effects on regulatory DNA sequence, using this representation to parameterize a conditional transport process that supports zero-shot inference at loci not seen during training [28]. This sequence-conditioned approach expands genomic coverage significantly compared to gene-level models, resolving functionally alternative transcription start sites that are often collapsed in standard representations. Additionally, frameworks like scBIG explicitly model coordinated gene programs through module-inductive representations, inducing coherent gene programs from data via Gene-Relation Clustering before applying conditional flow matching [17]. The convergence of diffusion models, flow matching, and optimal transport in these frameworks underscores a paradigm shift toward distributional, conditional, and biologically grounded generative modeling. By explicitly modeling the geometry of cellular state spaces and the dynamics of perturbation-induced transitions, these methods enable robust prediction of cellular responses in complex, unseen biological scenarios, laying the groundwork for truly predictive in silico experimentation and the realization of virtual cell technologies.

Integrating Biological Priors and Graph Structures

The transition from purely data-driven deep learning architectures to hybrid models that explicitly incorporate biological priors represents a critical evolution in single-cell foundation models. While early foundation models demonstrated proficiency in capturing gene expression patterns through self-supervised learning on massive corpora, they frequently treated genes as interchangeable tokens or independent samples, thereby ignoring the intricate regulatory topologies and taxonomic hierarchies inherent to biological systems [6]. Recent interpretability analyses have confirmed that while these models internalize organized biological knowledge such as pathway membership and protein interactions, they encode minimal causal regulatory logic, often failing to distinguish between direct regulatory targets and statistical co-expression [6]. To address these limitations, recent architectural innovations have focused on integrating Gene Regulatory Networks (GRNs), cell ontologies, and spatial hierarchies directly into the model structure, loss functions, and attention mechanisms. This integration aims to constrain the solution space to biologically plausible interactions, improve generalizability to unseen perturbations, and enhance the interpretability of learned representations.

A primary strategy involves embedding graph structures directly into the transformer architecture to model regulatory dependencies. Models such as GRNFormer and GenoHoption exemplify this approach by moving beyond sequence-based representations to leverage graph topological adapters. GRNFormer introduces a framework that systematically integrates multi-scale GRNs inferred from multi-omics data, utilizing a graph topological adapter with multi-head cross-attention to dynamically weight regulatory relationships [4]. Furthermore, it employs a novel edge perturbation strategy that augments graph neural network training with biologically informed co-expression links, addressing information asymmetry within the networks. Similarly, GenoHoption bridges the gap between foundation models and gene network graphs by introducing constraints that allow the model to learn predefined graph structures while expanding the receptive field without causing parameter explosion [21]. These methods contrast with standard attention mechanisms that may fail to capture long-range regulatory dependencies or distinguish between activation and inhibition without explicit structural guidance. In spatial contexts, this graph integration extends to modeling tissue microenvironments; for instance, HEIST models tissue as spatial cellular neighborhood graphs where each cell is further represented as a gene regulatory network graph, enabling hierarchical message passing that captures both local gene regulation and broader spatial context [81].

Beyond modifying the architecture, another significant avenue of research involves guiding the pre-training objective using biological ontologies. The Cell-ontology guided transcriptome foundation model (scCello) argues that treating cells as independent samples ignores critical taxonomic relationships available in cell ontology graphs [60]. To rectify this, scCello introduces cell-type coherence loss and ontology alignment loss alongside standard masked gene expression prediction. This forces the model to learn representations that respect the structural relations between cell types defined in the Open Biological and Biomedical Ontology Foundry, resulting in improved performance on tasks such as identifying novel cell types and predicting cell-type-specific marker genes. This approach aligns with broader arguments that foundation models in regulatory biology must be grounded in semi-mechanistic frameworks to achieve robust out-of-distribution performance, rather than relying solely on statistical correlations [76]. The integration of priors also extends to the inference of dynamic regulatory processes. Traditional methods often struggle with the static nature of single-cell snapshots, yet biological regulation is inherently temporal. Approaches like scPN utilize piecewise Ordinary Differential Equation (ODE) models to simultaneously infer pseudotime, velocity fields, and gene interaction networks, effectively reconstructing the dynamic connectivity matrices that govern cell differentiation [277]. Similarly, FLeCS incorporates gene network structures into coupled differential equations to model transcriptional dynamics at scale, allowing for the simulation of single-cell trajectories under perturbations without relying on low-dimensional embeddings [59]. These dynamical models address the "mean collapse" observed in perturbation prediction, where models predict global averages rather than specific responses, by explicitly modeling sparsity and biological structure through adaptive learning [70].

Despite the promise of integrating priors, challenges remain regarding the quality and completeness of the prior knowledge itself. Some methodologies, such as KINDLE, propose decoupling GRN inference from strict prior dependencies by distilling knowledge from a teacher model that uses priors into a student model that operates on expression data alone, thereby preserving the potential for novel biological discoveries [265]. Conversely, other

frameworks like EpiAwareNet utilize lightweight biological priors from bulk data as noisy positive edges to guide single-cell GRN reconstruction, demonstrating that even imperfect priors can enhance robustness when combined with adaptive cross-modal representation learning [130]. The tension between leveraging established knowledge to stabilize training and maintaining the flexibility to discover novel regulatory mechanisms defines the current landscape of architectural innovation. Furthermore, the application of these graph-structured and prior-guided models has shown tangible benefits in downstream tasks, including drug response prediction and spatial transcriptomics, where hierarchical graph transformers like HEIST model tissue as spatial cellular neighborhood graphs to capture microenvironmental influences [81]. Ultimately, the synthesis of graph structures, ontological constraints, and dynamical systems theory into foundation models marks a shift toward more mechanistic, interpretable, and biologically faithful computational frameworks.

Advances in Cell Type Annotation and Identity Understanding

Accurate cell type annotation serves as the cornerstone for deciphering cellular heterogeneity, yet the field is undergoing a fundamental paradigm shift from static, supervised classification to dynamic, reasoning-based frameworks powered by Large Language Models (LLMs) and foundation models. Traditional methods often falter when confronted with novel, rare, or long-tailed cell distributions that lack extensive reference labels, prompting the development of agentic systems that leverage semantic reasoning to automate and interpret annotation. Recent frameworks move beyond simple pattern matching to interpretative analysis, treating genes as tokens and cells as sentences to bridge the gap between raw expression data and biological context. For instance, [240] utilizes an LLM agent architecture to identify and discover novel cell types across diverse tissues, demonstrating superior extensibility compared to fixed-classification models. Similarly, [123] mimics expert practice by leveraging LLM-encoded knowledge to perform on-the-fly annotation with interpretable rationales, achieving significant accuracy gains over baselines like CellTypist, particularly in identifying rare and novel cell states. These agentic approaches are further refined by systems like [129], which integrates database verification to mitigate hallucinations, and [191], which employs multi-round debate among verifier agents to resolve ambiguous claims, providing auditable derivation trees rather than black-box predictions.

Despite the promise of these agentic and generative approaches, their reliability and biological grounding remain subjects of rigorous scrutiny, revealing a "baseline paradox" where complex models do not always outperform simpler statistical methods. Benchmarking studies such as [11] evaluate instruction-tuned LLMs across multiple datasets, finding that while these models provide robust interpretations without fine-tuning, their performance varies significantly depending on the task. More critical evaluations like [30] highlight substantial gaps, noting that even advanced generalist models often fail to outperform random guessing in complex tasks like drug response prediction, suggesting that current LLMs do not yet fully "understand" cell biology in a mechanistic sense. This sentiment is echoed in [186], which shows that current AI scientists substantially underperform human experts on discovery tasks, and [172], which argues for knowledge-augmented evaluation to overcome the brittleness of standard metrics. To address these reliability issues, researchers have developed strategies like [88], which establishes objective criteria to assess annotation credibility even in the absence of reference data, and [108], which employs reinforcement learning with batch-level rewards to solve reasoning puzzles, significantly outperforming off-the-shelf models by mimicking the contextual reasoning of human experts.

A critical advancement in addressing these limitations is the integration of semantic knowledge from natural language with transcriptomic data to create unified, biologically grounded representations. [57] introduces a framework that constructs unified representations of single-cell data and natural language, allowing for effective zero-shot cell identity understanding where traditional models struggle. This multimodal fusion is echoed in [scMMGPT: Language-Enhanced Representation Learning for Single-Cell Transcriptomics], which combines discriminative precision with generative flexibility, and [OKR-CELL: Open World Knowledge Aided Single-Cell Foundation Model with Robust Cross-Modal Cell-Language Pre-training], which uses retrieval-augmented generation to enrich cell descriptions with open-world knowledge. By treating genes as tokens, models like [12] and [44] bridge the gap between raw expression data and biological context, generating structured natural language descriptions that capture cellular identity more richly than discrete labels. Furthermore, [49] demonstrates that leveraging synergies between scGPT and LLM representations yields stronger, more consistent performance than either component alone, while [164] utilizes gene-specific textual annotations to generate biologically contextualized embeddings.

However, the reliance on massive datasets presents challenges regarding data efficiency and the handling of long-tailed distributions, necessitating specialized architectural innovations and semi-supervised strategies. Traditional foundation models often struggle with covariate shifts and rare cell populations. To counter this, [56] introduces a Gaussian Inflation Loss and Hard Data Mining strategy specifically designed to improve predictive accuracy for minority data samples. Complementary approaches like [61] utilize prototype-guided post-pretraining to refine latent embedding manifolds using marker-gene sets as structural priors. Additionally, semi-supervised frameworks such as [105] and [83] address the scarcity of labeled data by combining supervised learning for known types with unsupervised clustering to detect novel or out-of-domain cell types, providing uncertainty quantification and interpretable gene drivers. The evolution of these models also extends to integrating spatial and morphological contexts, recognizing that cell identity

is not defined solely by transcriptomics. Frameworks like [166] and [114] fuse spatial gene expression with histological images, while [21] incorporates gene network graphs to enhance the modeling of gene relationships. Collectively, these advances illustrate a trajectory toward more generalizable, interpretable, and context-aware annotation systems that transcend the limitations of static, supervised classifiers, moving toward a future where virtual cells and agentic reasoning drive biological discovery.

Zero-Shot Transfer and Handling Rare Cell Types

The paradigm of single-cell analysis is undergoing a critical re-evaluation regarding the efficacy of zero-shot transfer in foundation models. While large-scale pre-training promises generalization across datasets without extensive fine-tuning, recent rigorous benchmarking reveals a "baseline paradox" where simple, parameter-free linear representations often match or exceed complex foundation models on out-of-distribution tasks involving novel organisms [103]. This finding suggests that much of the biological signal in standard benchmarks resides in low-complexity manifolds accessible via careful normalization, yet it simultaneously highlights specific niches where advanced architectures are indispensable: handling extreme sparsity, resolving long-tailed cell type distributions, and bridging disparate modalities. To address the inherent noise and dropout rates exceeding 90% in single-cell data, recent innovations have moved away from reconstruction-based objectives, which often encode technical artifacts, toward latent predictive learning. For instance, Cell-JEPA introduces a joint-embedding predictive architecture that predicts cell-level embeddings from partial observations in latent space, leveraging the redundancy of cell identity encoding across genes to achieve a 36% relative improvement in zero-shot clustering compared to reconstruction-based baselines like scGPT [1]. By forcing models to learn dropout-robust features, such architectures provide the stability required to generalize to unseen cellular states, a prerequisite for identifying rare populations that are frequently underrepresented in training distributions.

The challenge of rare cell type identification is exacerbated by the long-tailed nature of cell-type distributions in large-scale atlases, where standard models bias representations toward abundant lineages. To mitigate this, specialized loss functions and data mining strategies have been developed to enhance sensitivity to minority classes. Celler incorporates a Gaussian Inflation Loss function that dynamically adjusts sample weights, improving the model's ability to learn from rare categories while preventing overfitting to common ones, coupled with a Hard Data Mining strategy that targets difficult minority samples [56]. Complementing this, CellRefine operates as a post-pretraining refinement stage, utilizing marker-gene sets as structural priors to guide the latent embedding manifold, effectively addressing covariate shifts and yielding gains of up to 15% over standard fine-tuning protocols for rare subtypes [61]. Furthermore, incorporating biological taxonomies directly into pre-training enhances zero-shot inference for novel types; scCello leverages cell ontology graphs during training via cell-type coherence and ontology alignment losses, allowing the model to learn structural relationships between cell types and improve the prediction of marker genes for unseen cells [60]. These approaches collectively suggest that incorporating biological priors and specialized loss functions is essential for unlocking the zero-shot potential of foundation models in the context of imbalanced data.

Beyond transcriptomics, robust zero-shot transfer increasingly relies on multimodal integration and external knowledge to handle novel contexts, such as de novo cell lines or cross-modal predictions. In the realm of image-based profiling, CellPainTR utilizes source-specific context tokens to achieve out-of-distribution generalization on entirely unseen datasets without fine-tuning, maintaining high performance despite significant domain shifts [205]. Similarly, for predicting gene expression from histopathology images, the PEKA framework employs parameter-efficient knowledge transfer to align image features with transcriptomic domains, achieving at least a 5% performance improvement over baseline foundation models by bridging the modality gap [133]. In drug response prediction, where identifying rare resistant subpopulations is critical, scAdaDrug employs importance-aware multi-source domain adaptation to identify drug-resistant cells across diverse patient cohorts, while efficient fine-tuning strategies using drug-conditional adapters enable zero-shot generalization to unseen cell lines by training less than 1% of the original model parameters [228; 5].

The frontier of handling rare and novel cell states is also being pushed by generative and agentic frameworks that simulate plausible cellular dynamics or reason over biological knowledge. Lingshu-Cell, a masked discrete diffusion model, serves as a cellular world model capable of generating realistic transcriptomic profiles for rare subtypes and predicting responses to novel perturbations by modeling the distribution of cellular states rather than static representations [32]. Concurrently, agentic systems like CellMaster leverage large language models to perform on-the-fly annotation with interpretable rationales, demonstrating particular strength in identifying rare and novel cell states where traditional baselines fail, especially when augmented with human-in-the-loop refinement [123]. Despite these advances, rigorous benchmarking remains essential to determine when such complex architectures are truly necessary versus when simpler methods suffice. Platforms like PyTDC and scSSL-Bench evaluate models across diverse modalities and therapeutic contexts, revealing that while foundation models excel in specific zero-shot and

multimodal tasks, their superiority is context-dependent and must be validated against robust, domain-specific metrics [50; 42]. Ultimately, the convergence of latent predictive learning, ontology-guided pre-training, and generative simulation represents a multifaceted strategy for overcoming data scarcity, ensuring that foundation models can effectively generalize to the full spectrum of cellular diversity.

LLM-Driven Annotation and Semantic Integration

The integration of instruction-tuned Large Language Models (LLMs) with single-cell transcriptomics represents a fundamental paradigm shift, moving the field from purely numerical pattern recognition to semantically grounded biological interpretation. Traditional foundation models often treat cells as independent vectors or sequences of gene tokens, frequently ignoring the rich taxonomic relationships and functional descriptions inherent in biological knowledge. To bridge this gap, emerging frameworks employ "text-bridge" strategies that initialize gene representations using descriptive text extracted from databases like NCBI, effectively injecting prior biological knowledge into the model architecture. For instance, [12] and [39] both leverage the general knowledge of models like LLaMA-2 to embed genes based on their functional descriptions before aggregating them into cell-level representations. Similarly, [164] demonstrates that ranking genes by expression and retrieving their textual annotations allows for the creation of compact, semantically rich embeddings that improve clustering and trajectory inference. These approaches transform abstract expression vectors into concepts that LLMs can reason about, addressing the limitation of models that rely solely on statistical co-expression patterns.

Beyond static initialization, the field is increasingly adopting agentic frameworks to address the subjectivity of cluster annotation and the complexity of resolution selection. Rather than relying on fixed classification heads, agents act as dynamic reasoning engines capable of handling context and novelty. [194] transforms annotation into an optimizable task by generating ranked Gene-Ontology hypotheses and evaluating their internal consistency and external distinctiveness, surpassing classical metrics like silhouette scores. This capability is extended by [240], which generalizes across tissues to discover novel cell types, and [191], which employs a multi-agent debate system to separate evidence grounding from label decisions. To further mitigate hallucinations, [129] integrates database verification directly into the agent workflow, while [123] mimics expert practice by leveraging LLM-encoded knowledge for zero-shot annotation with interpretable rationales. These agentic systems move beyond simple pattern matching, offering auditable reasoning traces that mimic expert curation and address the limitations of one-shot prompting observed in benchmarks like [108].

The synthesis of numerical data and semantic concepts is perhaps most evident in multimodal generative frameworks that translate expression profiles directly into natural language. [44] integrates gene-level embeddings with pretrained LLMs to generate coherent summaries of cellular identity and disease associations, providing interpretable outputs that discrete prediction heads cannot match. This generative capability is enhanced by [171], which pairs text commands with scRNA-seq profiles to enable complex tasks like conditional pseudo-cell generation through natural language. Furthermore, [57] addresses the lack of semantic understanding in single-modal models by constructing a unified representation of single-cell data and natural language during pre-training, enabling effective zero-shot cell identity understanding. These models suggest a future where researchers interact with data through conversation rather than code, facilitated by frameworks like [176] which unifies expression embeddings with semantic retrieval for interactive discovery.

However, the reliability and biological grounding of these LLM-based approaches remain subjects of intense scrutiny. Critical evaluations such as [30] reveal that while generalist models show preliminary understanding, many fail to significantly outperform random guessing in complex tasks like drug response prediction. This skepticism is reinforced by [103], which argues that much of the biological structure in current benchmarks is accessible through simple linear representations, urging the field to ensure that LLM-driven semantic integration captures genuinely novel insights rather than rediscovering known statistical structures. Conversely, [11] offers a more optimistic view, demonstrating that instruction-tuned LLMs can provide robust interpretations without fine-tuning. To navigate this tension, recent work focuses on objective evaluation, such as the "talk-to-machine" strategy proposed in [88], which establishes criteria for annotation reliability independent of expert bias. Ultimately, as surveyed in [84], the convergence of these approaches points toward a hybrid future where LLMs serve as both Oracles for direct modeling and Agents for orchestrating discovery, provided their outputs are rigorously validated against biological reality.

Robustness to Batch Effects and Out-of-Distribution Detection

The reliability of automated cell type annotation in single-cell transcriptomics is fundamentally contingent upon a model's ability to distinguish biological signal from technical noise and to recognize when input data deviates from the training distribution. Technical batch effects, arising from variations in sample preparation, sequencing platforms, or processing pipelines, often introduce systematic biases that can obscure genuine biological heterogeneity or create artificial clusters [49]. Consequently, robust annotation frameworks must achieve invariance to these technical artifacts while maintaining high sensitivity to novel cellular states that fall outside the scope of reference atlases. Recent advancements have addressed these dual challenges through diverse architectural innovations, ranging from spiking neural networks and evidential learning to generative diffusion models and agent-based reasoning systems.

A primary strategy for enhancing robustness involves modifying the underlying neural architecture to better handle sparse, noisy, and batch-confounded data. SpikGPT introduces a hybrid deep learning framework that integrates scGPT-derived cell embeddings with a spiking Transformer architecture, specifically designed to maintain accuracy in the presence of strong batch effects [64]. By employing a multi-head Spiking Self-Attention mechanism, SpikGPT achieves energy-efficient feature extraction that matches or exceeds leading annotation tools across multiple benchmarks. Uniquely, this framework addresses the out-of-domain (OOD) challenge by assigning low-confidence predictions to an "Unknown" category, thereby allowing for the accurate rejection of cell states absent from the training reference without requiring explicit OOD training samples [64]. Similarly, CellPainTR tackles batch effects in image-based profiling by utilizing source-specific context tokens within a Transformer architecture. This design enables effective OOD generalization to entirely unseen datasets without fine-tuning, outperforming traditional batch integration methods like ComBat and Harmony in preserving biological signals while mitigating technical variation [205]. In the realm of efficiency and scalability, GeneMamba incorporates biologically meaningful loss functions and State Space Models to process over 50 million cells, demonstrating strong performance in multi-batch integration where traditional transformers struggle with quadratic complexity [16].

Parallel to architectural modifications, probabilistic approaches have emerged as a rigorous method for quantifying uncertainty and detecting OOD cells. The eDOC framework leverages evidential learning within a Transformer decoder to simultaneously annotate in-domain cells and identify OOD populations by modeling the uncertainty of predictions [83]. Unlike methods that rely on maximum softmax probabilities, eDOC quantifies the reliability of annotations and highlights specific gene drivers that contribute to a cell being classified as OOD, offering interpretability at single-cell resolution [83]. This capability is critical for precision medicine, where identifying novel disease-associated cell types is as important as classifying known ones. Complementing this, PRESCRIBE employs a multivariate deep evidential regression framework to jointly estimate epistemic and aleatoric uncertainty in perturbation predictions, ensuring that confidence scores correlate strongly with empirical accuracy [151]. These evidential approaches provide a theoretical grounding for rejecting unreliable predictions, a feature often lacking in standard supervised classifiers.

Generative models and self-supervised learning strategies offer alternative pathways to robustness by learning latent representations that are inherently resistant to noise and distribution shifts. Cell-JEPA shifts the learning objective from reconstructing sparse counts to predicting cell-level embeddings in latent space, forcing the model to learn dropout-robust features that capture redundant cellular identity information [1]. This approach yields significant improvements in zero-shot transfer performance compared to reconstruction-based models like scGPT. Furthermore, scDiVa and Lingshu-Cell utilize masked discrete diffusion mechanisms to model the distribution of transcriptomic states, aligning the generation process with the dropout-like corruption inherent in scRNA-seq data [43][32]. By operating in discrete token spaces and avoiding the artificial ordering biases of autoregressive models, these frameworks demonstrate strong transfer performance in batch integration and OOD scenarios. Set-DINO and WS-DINO further illustrate the power of weakly supervised and self-supervised learning in morphological profiling, leveraging replicate structures to mitigate confounders and encode biologically meaningful information despite batch variations [125][255].

The integration of large language models (LLMs) and agentic frameworks introduces a new paradigm for handling OOD detection through reasoning and external knowledge retrieval. CellMaster and scAgent utilize LLM-encoded knowledge to perform zero-shot annotation, demonstrating particular strength in identifying rare and novel cell states where baseline models often fail due to rigid reference constraints [123][240]. These systems mimic expert practice by

evaluating marker genes and tissue context dynamically, rather than relying solely on fixed embedding spaces. MAT-Cell advances this by employing a multi-agent tree-structured reasoning framework that separates evidence grounding from label decisions, using debate mechanisms to challenge claims before reaching a consensus, which proves effective when target states are poorly covered by reference atlases [191]. Additionally, Cell-o1 trains LLMs via reinforcement learning with batch-level rewards to solve "CellPuzzles," enforcing reasoning across cellular context to ensure label uniqueness and improve generalization across diverse donor conditions [108].

Despite these advances, contradictions and limitations persist regarding the interpretability of attention mechanisms and the generalizability of foundation models. While some studies suggest that attention scores in Transformer models can highlight marker genes, systematic evaluations indicate that attention patterns often capture co-expression rather than unique regulatory signals, limiting their utility for mechanistic interpretation in OOD cases [8]. Furthermore, while models like scGPT and Geneformer show promise, their performance can degrade under significant domain shifts without specific adaptations such as the post-pretraining strategies proposed by CellRefine or the conformal inference constraints used in scConform to ensure prediction set coherence [61][268]. The field is thus moving towards hybrid approaches, such as OKR-CELL and scMMGPT, which combine cross-modal pretraining with robust alignment objectives to leverage open-world knowledge while resisting noise, aiming to create annotation tools that are both universally applicable and rigorously uncertain about their own limitations [224][24].

Modeling and Predicting Cellular Perturbation Responses

Predicting cellular responses to genetic and chemical perturbations represents a cornerstone of modern drug discovery, necessitating a fundamental shift from static profiling toward dynamic simulations of cellular states. A central challenge driving recent architectural innovations is the "Perturbation Suppression Hypothesis," which posits that generic training objectives often fail to capture sparse, perturbation-specific signals because they are overwhelmed by dominant, perturbation-invariant biological programs [185]. This imbalance leads to models that either entangle invariant background information with perturbation effects or suppress the specific signals required for accurate prediction. Empirical analyses reveal that representations learned by large-scale foundation models often exhibit weaker linear decodability of perturbation labels compared to simple baselines like Principal Component Analysis, suggesting that generic pre-training objectives prioritize shared cellular structures over intervention-specific changes [185]. Consequently, the field is moving beyond simple reconstruction tasks to frameworks that explicitly disentangle these factors, isolate causal mechanisms, and model full distributional shifts.

To address the suppression of sparse signals, novel frameworks have been proposed to explicitly separate perturbation-specific information from invariant structures. For instance, PerturbedVAE utilizes alignment-based components and latent causal models to recover sparse signals, achieving state-of-the-art performance in out-of-distribution combinatorial predictions [185]. Similarly, the Sparse Additive Mechanism Shift Variational Autoencoder (SAMS-VAE) models perturbed states as a sum of local sample-specific variations and sparse global intervention effects, enabling compositional reasoning for unseen combinatorial perturbations [201]. These approaches contrast with methods that rely solely on dense representations, which often suffer from "mean collapse," where predictions regress toward the global average expression rather than capturing true differential effects [70]. ADAPERT further mitigates this by learning perturbation-specific subgraphs from biological knowledge graphs, separating true signals from noise and avoiding the error propagation common in static graph models [70].

Beyond disentangling representations, there is a growing consensus that modeling the full distribution of cellular responses, rather than just mean shifts, is critical for capturing biological heterogeneity. Traditional methods often assume cell-level correspondences that do not exist due to the destructive nature of single-cell sequencing, leading to inaccurate mappings. Generative frameworks based on conditional flow matching, such as scDFM and CFM-GP, model the transformation between control and perturbed distributions, effectively capturing population-level shifts and higher-order statistics like variance and skewness [19; 52]. Diffusion models have also emerged as powerful tools for this task; PerturbDiff operates directly over probability distributions in a Hilbert space to account for unobservable latent factors, while Departures approximates Schrödinger Bridges to align control and perturbed populations without requiring paired data [45; 71]. These distributional approaches significantly outperform baselines in combinatorial settings and provide a more realistic view of stochastic cellular responses [149].

The integration of biological priors and mechanistic understanding further enhances model robustness and interpretability, addressing the limitations of purely data-driven approaches. Semi-mechanistic models argue that incorporating experimental design and biological constraints is essential for improving out-of-distribution generalization [76]. For instance, STRAND conditions perturbation effects on regulatory DNA sequences rather than fixed gene identifiers, enabling zero-shot inference at loci unseen during training and expanding genomic coverage significantly [28]. In the realm of temporal dynamics, models like PerturbODE and FLeCS utilize neural ordinary differential equations to infer gene regulatory networks and simulate cell state trajectories, capturing the continuous evolution of cellular programs under intervention [235; 59]. Despite these advances, benchmarking efforts indicate that no single architecture universally dominates, and simpler models often remain competitive when scaled appropriately [47]. Some studies even suggest that classical statistical methods or linear models can outperform complex deep learning architectures in specific transcriptomic prediction tasks, highlighting the need for rigorous evaluation standards that go beyond traditional fit measures like RMSE to include rank metrics and biological consistency [31; 47]. The field is increasingly recognizing the value of multi-modal integration, where models leverage morphological and transcriptomic data jointly to infer causal links and predict responses across different measurement modalities [188; 298]. The development of "virtual cell" world models aims to unify these capabilities into interpretable, stepwise simulators that can reason about signaling cascades and developmental trajectories, bridging the gap between black-box predictions and mechanistic biological understanding [237; 137].

Conclusion

The field of single-cell genomics stands at a critical inflection point, transitioning from an era defined by static, task-specific deep learning to a new paradigm of dynamic, generative foundation models capable of simulating cellular "worlds." This evolution represents more than a mere scaling of parameters; it is a fundamental reimagining of how computational biology captures the complexity of life. By treating genes as tokens and cells as sentences, modern architectures have begun to decode the context-dependent syntax of gene regulation, enabling zero-shot transfer learning and the integration of multimodal data ranging from spatial transcriptomics to proteomics. Yet, this rapid advancement is tempered by a profound and recurring contradiction known as the "baseline paradox," which challenges the necessity of massive model complexity. Rigorous benchmarking consistently reveals that simple, parameter-free linear representations often match or exceed the performance of state-of-the-art foundation models on standard downstream tasks. This phenomenon suggests that current evaluation metrics may inadvertently reward the capture of dominant, perturbation-invariant background signals while failing to isolate the sparse, causal mechanisms that drive specific cellular responses.

Consequently, the field faces a pivotal realization: scaling parameters alone is insufficient for achieving true biological understanding. Without explicit constraints, deep learning architectures risk becoming sophisticated engines of statistical co-expression rather than tools for mechanistic discovery, capturing correlation while missing the underlying regulatory grammar. The future trajectory of single-cell foundation models must therefore converge on hybrid, semi-mechanistic frameworks that marry the scalability of data-driven learning with the rigor of causal inference and biological priors. Achieving the vision of a "virtual cell" requires architectures that explicitly disentangle invariant cellular programs from perturbation-specific effects, integrating gene regulatory networks and dynamical systems theory directly into the learning process. Simultaneously, the emergence of agentic AI offers a pathway to transcend the limitations of passive pattern recognition, enabling systems that can reason over biological literature, automate hypothesis generation, and validate predictions against interventional data. Ultimately, the transition from powerful descriptive tools to genuine engines of discovery depends on shifting the focus from optimizing predictive accuracy on static benchmarks to developing robust, causally grounded models capable of reliable simulation and explanation in the face of biological complexity.